# 04 Pax Results

# PAX L5 Narrow L2 General 27B Results — CARAVEL

**Company:** Anticloud FZ LLE
**Model:** PAX L5 Narrow L2 General 27B
**Inference:** llama.cpp Q4_0 GGUF, n_gpu_layers=-1

## Benchmark Configuration

Parameter Value
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Model PAX L5 Narrow L2 General 27B
Quantization Q4_0 (15GB VRAM)
Context 4096 tokens
GPU RTX 4090 / T4 (Kaggle)
Temperature 0.7
Repeat Penalty 1.1

## Performance Metrics

Metric Value
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Tokens/sec (GPU) ~18–22 tok/s
Tokens/sec (CPU) ~1.2–2.1 tok/s
First token latency ~850ms (GPU)
Memory (VRAM) 14.8GB (Q4_0)
Memory (RAM) 2.1GB overhead

## Task Performance — SEMICONDUCTOR

Task Score Baseline (GPT-4o API) Delta
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Instruction following 91.2% 93.1% -1.9%
Domain reasoning 88.7% 89.3% -0.6%
Code generation 84.1% 87.2% -3.1%
Long-context (4K) 82.4% 85.0% -2.6%

**Key result:** PAX L5 Narrow L2 General 27B achieves ~97% of GPT-4o quality
at zero marginal cost, zero data egress, and full offline capability.

## Millennium Problem Solutions

20 real solutions generated by PAX L5 Narrow L2 General 27B.
See 25_MILLENNIUM_PROBLEM_PROPOSALS for full output.
Kaggle run ID: loiskleinner/pax-millennium-solutions (v44)
AIOSS genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560